

AMENDMENT NO. 1 NOVEMBER 1978
TO
IS : 8496-1977 SPECIFICATION FOR
PROPOXUR, TECHNICAL

Alteration

(*Pages 5 and 6, Appendix A*) — Substitute the following for the existing matter:

A P P E N D I X A
[*Table 1, Item (i)*]

DETERMINATION OF PROPOXUR CONTENT

A-0. GENERAL

A-0.1 For the determination of propoxur content, two methods have been prescribed. Either of these can be used as routine method.

A-1. HYDROLYSIS METHOD

A-1.1 Principle — The sample is hydrolysed with 60 percent sulphuric acid under reflux and the methylamine formed is distilled into 2 percent boric acid solution and titrated directly against standard 0.1 N sulphuric acid using bromocresol green indicator.

A-1.2 Reagents

A-1.2.1 Sulphuric Acid Solution — 60 percent (*m/v*). Prepare by adding 33 ml of concentrated sulphuric acid to 50 ml of distilled water. Mix and allow to cool. Make up to 100 ml with distilled water.

A-1.2.2 Boric Acid Solution — 2 percent (*m/v*). Prepare by dissolving 20 g boric acid in about 500 ml of hot water, cool, add 2 ml of 0.1 percent bromocresol green, dilute to one litre, and mix. This solution need not be prepared accurately and may be prepared in a large volume sufficient for several weeks' use. If the colour fades on standing, more indicator may be added to the remaining portion.

A-1.2.3 Sodium Hydroxide Solution Concentrated — Add 40 g of sodium hydroxide to about 75 ml of distilled water. Mix and allow to cool. Make up to 100 ml with distilled water.

A-1.2.4 Bromocresol Green Indicator Solution — 1 percent (*m/v*). Dissolve bromocresol green in ethanol, alternatively prepare an aqueous solution of the sodium salt as follows.

A-1.2.4.1 Rub 100 mg of bromocresol green in an agate mortar with 2.9 ml of 0.05 N sodium hydroxide. After the indicator has dissolved completely, add sufficient water to make the volume up to 100 ml.

A-1.3 Procedure — Weigh accurately about 0.8 g of the sample into a 250-ml round bottomed flask fitted with a ground-glass stopper, add 50 ml of sulphuric acid solution, heat under reflux for one and a half hours, then add 10 ml of sulphuric acid solution and reflux for another half an hour. Cool and transfer the digested sample quantitatively to a one-litre flask. Add 600 ml of the distilled water and 100 ml of concentrated sodium hydroxide solution. Immediately connect the flask to a condenser (all glass) and distil the liberated methylamine into 750-ml flask containing 50 ml of boric acid solution and bromocresol indicator, until 300 to 350 ml of liquid are collected. Remove the receiver and titrate the contents with 0.1 N standard sulphuric acid until the solution has been brought back to its original yellow-green colour.

A-1.4 Calculation

$$\text{Propoxur content, percent by mass} = \frac{V N \times 20.9}{M}$$

where

V = volume, in ml, of 0.1 N standard sulphuric acid required to titrate the steam-distilled methylamine;

N = normality of standard sulphuric acid; and

M = mass, in g, of sample taken for the test.

A-2. SAPONIFICATION METHOD

A-2.1 Principle — Propoxur is saponified with potassium hydroxide in diethylene glycol and the methyl amine formed is directly absorbed in boric acid solution and titrated against standard hydrochloric acid.

A-2.2 Apparatus — Assemble the apparatus as shown in Fig. 1.

A-2.3 Reagents

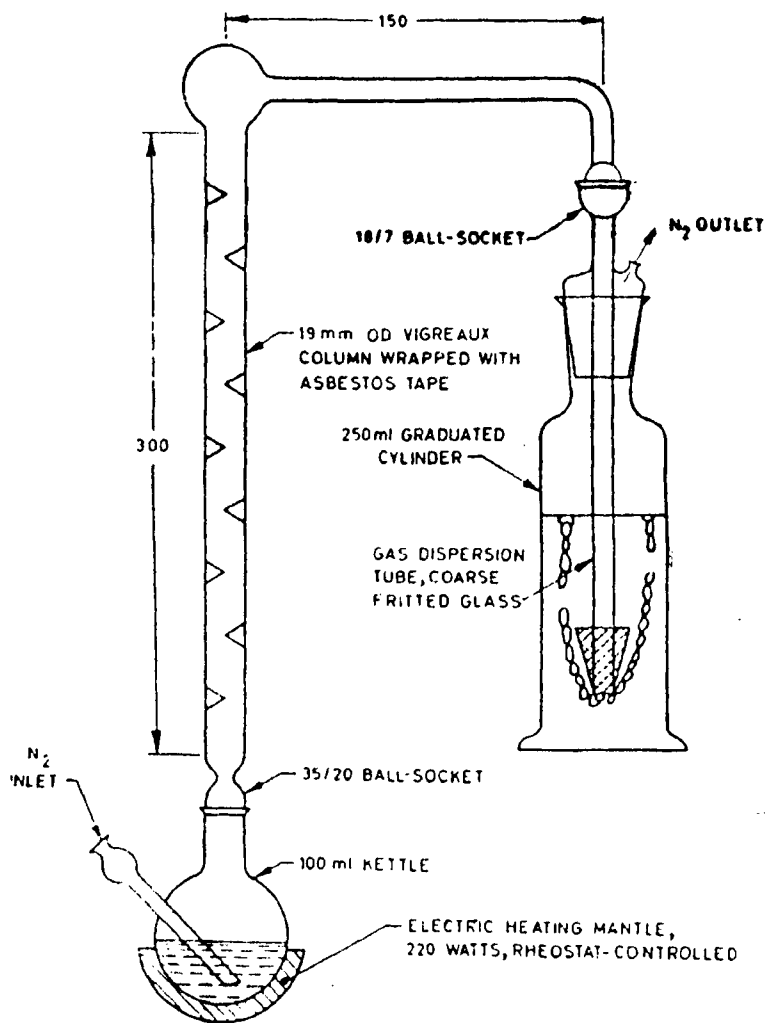
A-2.3.1 Potassium Hydroxide in Diethylene Glycol — 2 N. Dissolve 56 g of potassium hydroxide in a minimum quantity of water and make up to 500 ml with diethylene glycol.

A-2.3.2 Boric Acid Solution — See A-1.2.2.

A-2.3.3 Nitrogen

A-2.3.4 Bromocresol Green Indicator Solution — See A-1.2.4.

A-2.3.5 Hydrochloric Acid — 0.1 N.



All dimensions in millimetres.

FIG. 1 APPARATUS FOR DETERMINATION OF PROPOXUR CONTENT

A-2.4 Procedure — Weigh accurately about 0.5 g of propoxur in the 100-ml reaction flask. Add 50 ml of potassium hydroxide in diethylene glycol and connect the flask to the evolution assembly and nitrogen. Place 100 ml of boric acid solution in the 250-ml measuring cylinder and connect the same to the gas dispersion tube. Connect nitrogen from

cylinder and adjust the flow rate to 150 ml/minute. Start heating the flask and attain boiling within 5 minutes. Continue heating and passing nitrogen for a period of one and a half hours. Disconnect the cylinder containing boric solution and transfer into a 250-ml conical flask. Wash the cylinder and the dispersion tube with distilled water and add the washings to the contents of the conical flask. Titrate immediately against 0.1 N hydrochloric acid. The end point is from green to yellow.

A-2.5 Calculation

$$\text{Propoxur content, percent by mass} = \frac{V \times N \times 20.9}{M}$$

where

V = volume of standard hydrochloric acid consumed in the test;

N = normality of hydrochloric acid; and

M = mass of sample, in g, taken for the test.

(AFCD 6)